

Project Logbook — MediMatch

Prepared on: 2025-12-12

- Student name: _____
- Student ID: _____
- Programme/Course: _____
- Supervisor: _____
- Project title: MediMatch — Medical Supplies Matching Platform
- Period covered: Weeks 1–2 filled; Week 3+ template included

Supervisor Acknowledgement (cover):

- Name: _____
- Signature: _____ Date: _____

Week 1

- Dates: 2025-09-29 → 2025-10-03

Quick fill (optional):

Field	Entry
Goals (top 1–3)	_____
Evidence/Links	_____
Hours	_____
Notes	_____
Supervisor initials	_____

- Objectives:
 - Establish codebase scaffolding for client, server, and shared types.
 - Implement core auth and listings endpoints.
 - Provide an initial React UI with basic flows (Auth, Dashboard, Home).
- Activities:
 - Initialized monorepo with workspaces: `server/`, `client/`, `shared/`.
 - Implemented API routes: auth (register/login/me), listings (list/create), matches (scaffold), health.
 - Set up data backends: mock (in-memory), file DB (JSON), Postgres/PostGIS (configurable).
 - Server: TypeScript, `ts-node-dev`, `dotenv`, `jwt`, `bcrypt`.
 - Client SPA (React + Vite + TS) with Home, Auth, Dashboard; protected routes.
 - Shared TypeScript interfaces for `User` and `Listing`.

- Deliverables/Evidence:
 - Running dev environment (server + client), README instructions, baseline UI.
- Validation/Testing:
 - TypeScript build checks PASS (server).
 - Tests PASS (4/4) with mock DB; manual smoke tests for register/login/create listing.
- Time spent (hrs): 8–10 (setup, endpoints, UI, tests).
- Challenges/Risks:
 - None critical; Postgres optional for MVP.
- Next steps:
 - Polish UI/UX; assets/branding; one-command developer experience.

Supervisor comments (Week 1):

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Week 2

- Dates: 2025-10-06 → 2025-10-10

Quick fill (optional):

Field	Entry
Goals (top 1–3)	_____
Evidence/Links	_____
Hours	_____
Notes	_____
Supervisor initials	_____

- Objectives:
 - Make the app presentation-ready (polished UI/UX for Home/Auth/Dashboard).
 - Add media assets and branding.
 - Ensure a one-command developer experience.
- Activities:
 - Server/dev experience: `USE_FILE_DB=true` for local, API at `http://127.0.0.1:4000`.
 - One-command startup via root scripts: `predev` (sync assets), `dev` (concurrently run), `up`.
 - Client/UI/UX: hero background with gradient, glass overlay card, CTA; feature cards; auth form polish; dashboard empty state; styles in `client/src/styles.css`.
 - Assets integrated (icons and photos) via sync script to `client/public/images`.
 - Config: `VITE_API_URL` support; updated `tsconfig` and `favicon/title`.
- Deliverables/Evidence:
 - Polished Home/Auth/Dashboard UI, assets in repo, start scripts.
- Validation/Testing:
 - Client dev server PASS; server health PASS; tests PASS (4/4); manual UX walkthrough PASS.
- Time spent (hrs): 10–12 (UX polish, assets, scripts, validation).
- Challenges/Risks:
 - Minor `npm audit` items; a benign deprecation warning; deferred.
- Next steps:
 - Prototype matching `/api/matches/suggest`; add input validation; toasts; expand tests.

Supervisor comments (Week 2):

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Week 3 (Template)

- Dates: 2025-10-13 → 2025-10-17

Quick fill (optional):

Field	Entry
Goals (top 1–3)	Matching prototype; client Suggested tab; input validation
Evidence/Links	/api/matches/suggest; client Dashboard Suggested tab
Hours	10–12
Notes	Haversine fallback for non-PostGIS; simple filters (category/radius)
Supervisor initials	_____

- Objectives:
 - Implement matching prototype using haversine distance and simple weighting.
 - Add /api/matches/suggest endpoint and unit tests.
 - Add client “Suggested” tab with basic filters and toasts.
- Activities:
 - Server: added matching utility; route/controller for suggest; manual guards for query params.
 - Client: new Suggested tab; radius/category filters; lightweight result cards; success/error toasts.
 - Updated API service to accept VITE_API_URL base and new endpoint.
- Deliverables/Evidence:
 - Endpoint: GET /api/matches/suggest?
lat=..&lon=..&radiusKm=..&limit=...
 - Tests: vitest coverage for input guards and ranking order.
 - UI: Suggested tab with filter controls and results list.
- Validation/Testing:
 - Unit tests PASS; manual checks with seeded sample data; edge cases for missing/invalid lat/lon handled.
- Time spent (hrs): 10–12
- Challenges/Risks:
 - Geolocation availability and accuracy; mitigated with manual lat/lon inputs.
- Next steps:
 - Integrate PostGIS for accurate distance and indexing; prepare migrations.

Supervisor comments (Week 3):

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Week 4 (Template)

- Dates: 2025-10-20 → 2025-10-24

Quick fill (optional):

Field	Entry
Goals (top 1–3)	Postgres/PostGIS setup; migrations; distance via ST_Distance
Evidence/Links	db/docker-compose.yml; server/src/tools/migrate.ts
Hours	8–10
Notes	Fallback to file DB maintained for dev
Supervisor initials	_____

- Objectives:
 - Provision Postgres + PostGIS via Docker and run schema migrations.
 - Add spatial index and geography(Point,4326) column for listings.
 - Update matching to use ST_Distance when PostGIS is available.
- Activities:
 - Authored idempotent migration script; enabled PostGIS extension; created GiST index.
 - Wired server config to detect PostGIS and switch distance strategy.
 - Updated docs/runbook; added Adminer note for DB inspection.
- Deliverables/Evidence:
 - Migrations (server/src/tools/migrate.ts); PostGIS init (db/init/postgis.sql).
 - Verified spatial query path and fallback path via logs/tests.
- Validation/Testing:
 - Ran `docker compose up -d` for DB; executed migrations; manual query checks for ST_Distance.
- Time spent (hrs): 8–10
- Challenges/Risks:
 - Windows Docker resource limits; tuned WSL integration and memory.
- Next steps:
 - Seed data for realistic spatial distribution; continue performance checks.

Supervisor comments (Week 4):

- Comments: _____
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Week 5 (Template)

- Dates: 2025-10-27 → 2025-10-31

Quick fill (optional):

Field	Entry
Goals (top 1–3)	Messaging (conversations/messages); Chat modal; REST flows
Evidence/Links	<code>server/src/controllers/messages.controller.ts</code> ; ChatModal UI
Hours	9–11
Notes	REST polling for simplicity; sockets deferred
Supervisor initials	_____

- Objectives:
 - Implement conversations/messages models and REST endpoints.
 - Build client Chat modal integrated with listings/users.
- Activities:
 - Server: routes/controllers for conversations list/create and message send/list; basic guards.
 - Client: ChatModal wired from listing detail; unread badge concept stubbed.
 - Tests: integration tests for message creation/listing.
- Deliverables/Evidence:
 - Endpoints under `/api/messages` and `/api/chat/*` (per routing structure).
 - UI demo showing send/receive via polling.
- Validation/Testing:
 - Supertest flows PASS; manual two-user conversation verified.
- Time spent (hrs): 9–11
- Challenges/Risks:
 - Real-time complexity; decision to keep polling for MVP.
- Next steps:
 - Add minimal unread counts; consider WebSocket upgrade later.

Supervisor comments (Week 5):

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Week 6 (Template)

- Dates: 2025-11-03 → 2025-11-07

Quick fill (optional):

Field	Entry
Goals (top 1–3)	Ratings & reputation; Notifications plumbing
Evidence/Links	<code>ratings.controller.ts</code> ; <code>notifications.controller.ts</code> ; RatingModal
Hours	8–10
Notes	Aggregates recalculated on write; simple notification types
Supervisor initials	_____

- Objectives:
 - Allow users to rate others; compute basic reputation aggregates.
 - Implement in-app notifications and listing of unread items.
- Activities:
 - Server: ratings create/list; uniqueness/limits; aggregate updates.
 - Server: notifications model + list/mark-as-read endpoints.
 - Client: RatingModal and badge indicators for notifications.
- Deliverables/Evidence:
 - Endpoints for ratings and notifications; UI integration points.
- Validation/Testing:
 - Unit tests for aggregates; manual test of rating flow and unread counters.
- Time spent (hrs): 8–10
- Challenges/Risks:
 - Preventing rating spam; enforced one-per-relationship guard.
- Next steps:
 - Expand notification types (match suggestions, new messages).

Supervisor comments (Week 6):

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Week 7 (Template)

- Dates: 2025-11-10 → 2025-11-14

Quick fill (optional):

Field	Entry
Goals (top 1–3)	Admin dashboard; user verification; listing moderation
Evidence/Links	server/src/routes/admin.routes.ts; pages/Admin.tsx
Hours	9–11
Notes	Guarded by authMiddleware + requireAdmin
Supervisor initials	_____

- Objectives:
 - Provide admin endpoints for stats, users, and listings moderation.
 - Add verification and disable flags management.
- Activities:
 - Implemented admin routes (stats, users list/update, listings list/hide/delete).
 - Added safeguards: cannot demote/disable self; cannot remove last active admin; verify org fields present.
 - Client admin page to view and act on users/listings.
- Deliverables/Evidence:
 - Admin endpoints including /api/admin/stats, /api/admin/users, /api/admin/listings, and reports.
- Validation/Testing:
 - Manual admin flows; unit tests for protected route guards.
- Time spent (hrs): 9–11
- Challenges/Risks:
 - Avoiding lockout and ensuring last-admin protection.
- Next steps:
 - CSV/JSON export for admins; refine UI batch actions.

Supervisor comments (Week 7):

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Week 8 (Template)

- Dates: 2025-11-17 → 2025-11-21

Quick fill (optional):

Field	Entry
Goals (top 1–3)	Security hardening; JWT cookies; CORS/Helmet/rate limit
Evidence/Links	security middleware config; auth cookie settings
Hours	8–9
Notes	Cache-control no-store on API; input validation tightened
Supervisor initials	_____

- Objectives:
 - Harden security: Helmet, CORS allowlist, rate limiting, stricter validation.
 - Move JWT to HttpOnly cookies for improved handling.
- Activities:
 - Added Helmet/CORS/rate-limit; adjusted cookie flags; refined request validation.
 - Updated docs for environment variables and origins.
- Deliverables/Evidence:
 - Middleware configuration; updated auth flows and docs.
- Validation/Testing:
 - Manual verification of cookie flags; negative-path tests for validation.
- Time spent (hrs): 8–9
- Challenges/Risks:
 - Balancing developer convenience with stricter CORS/rate limits.
- Next steps:
 - Pen-test checklist and dependency updates.

Supervisor comments (Week 8):

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Week 9 (Template)

- Dates: 2025-11-24 → 2025-11-28

Quick fill (optional):

Field	Entry
Goals (top 1–3)	Observability endpoints; admin reports CSV/JSON
Evidence/Links	<code>/api/admin/reports/summary(.csv)</code> ; <code>/api/health</code>
Hours	7–9
Notes	Category breakdown and 7/30-day metrics implemented
Supervisor initials	_____

- Objectives:
 - Add health/readiness/stats endpoints and admin reporting (JSON/CSV).
- Activities:
 - Implemented summary queries (users/listings/matches/messages, last 7/30 days, categories).
 - CSV generator with safe value sanitization; content-disposition headers.
- Deliverables/Evidence:
 - Endpoints: `/api/admin/reports/summary` and `/api/admin/reports/summary.csv`.
 - Example CSV exported for review.
- Validation/Testing:
 - Manual verification of counts vs database; CSV downloaded and opened successfully.
- Time spent (hrs): 7–9
- Challenges/Risks:
 - Query performance on large datasets; mitigated with indexes and simple aggregations.
- Next steps:
 - Pagination for large admin lists; scheduled exports (cron) later.

Supervisor comments (Week 9):

- Comments: _____
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Week 10 (Template)

- Dates: 2025-12-01 → 2025-12-05

Quick fill (optional):

Field	Entry
Goals (top 1–3)	QA hardening; docs finalization; demo prep
Evidence/Links	Chapter 7 updates; logbook PDF; runbook scripts
Hours	8–10
Notes	Focus on polish, tests, and submission assets
Supervisor initials	_____

- Objectives:
 - Stabilize MVP, improve tests and documentation, and prepare demo.
- Activities:
 - Expanded negative-path tests; refreshed screenshots; updated runbook and quick-start.
 - Image optimization and minor UI polish; resolved small bugs.
- Deliverables/Evidence:
 - Updated docs (Chapter 7, logbook); final build instructions; demo outline.
- Validation/Testing:
 - All server tests PASS; manual E2E walkthrough PASS.
- Time spent (hrs): 8–10
- Challenges/Risks:
 - Timeboxing scope; deferred advanced features to future work.
- Next steps:
 - Post-submission retrospective and backlog grooming.

Supervisor comments (Week 10):

- Comments: _____
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- Student signature: _____ Date: _____

Appendix (Artifacts/References)

- docs/Logbook_Week1-2.md (source notes for Weeks 1–2)
- docs/Chapter_1_and_Logbook_Update.md (intro, plan, weekly notes)

- docs/CHAPTER_7_IMPLEMENTATION.md (implementation details)
- Quick run (file DB): `npm run up`
- Postgres dev: `cd db; docker compose up -d`